June 2023 Hundred-Patch Benchmark

Metrics, plots, Taylor diagrams, optimization behavior, and runtimes

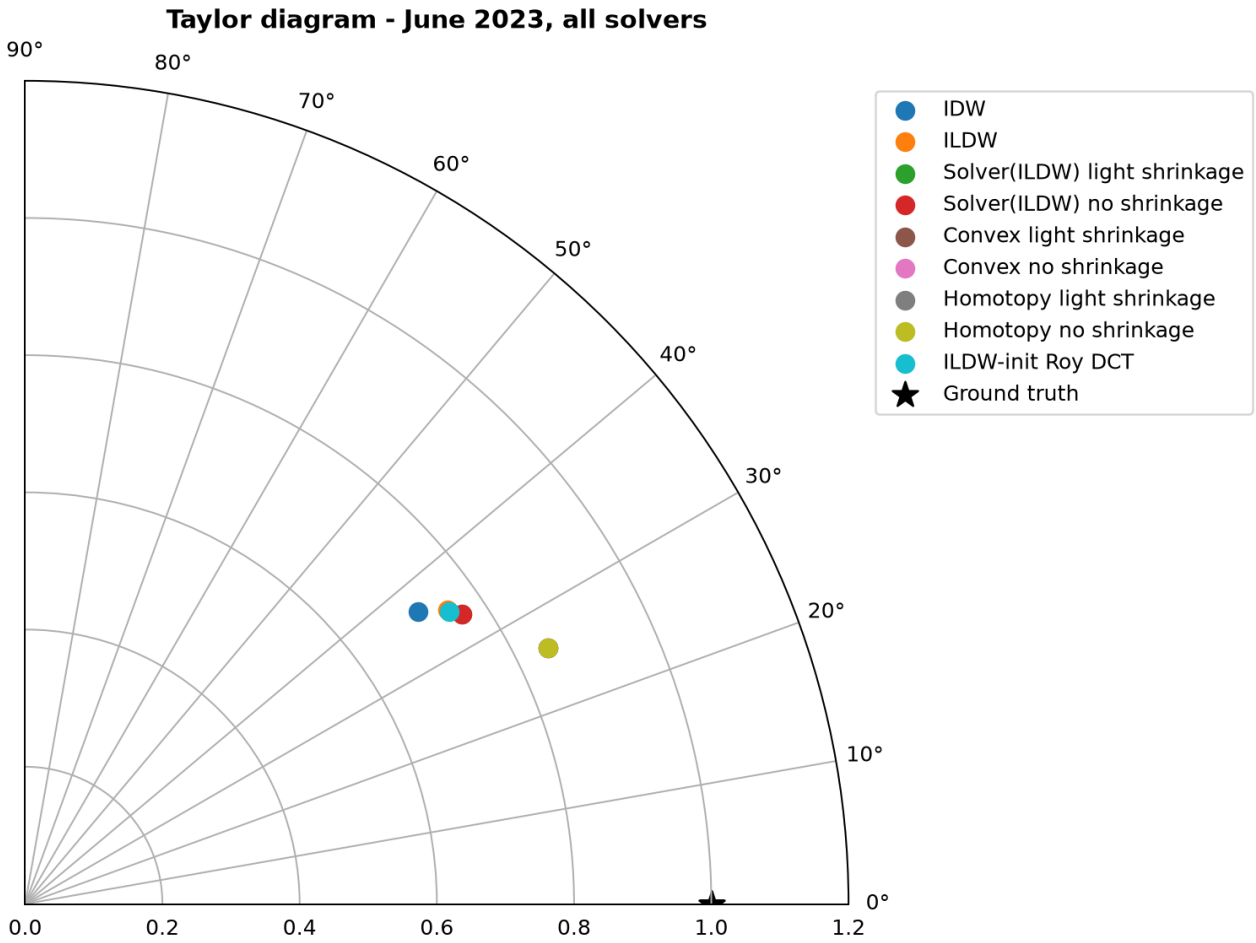
100 June 2023 patches. Supplementary Solver(ILDW) no-shrinkage, timed Solver(ILDW), and Roy DCT outputs are included from their complete June solution directories.

1. Summary metrics

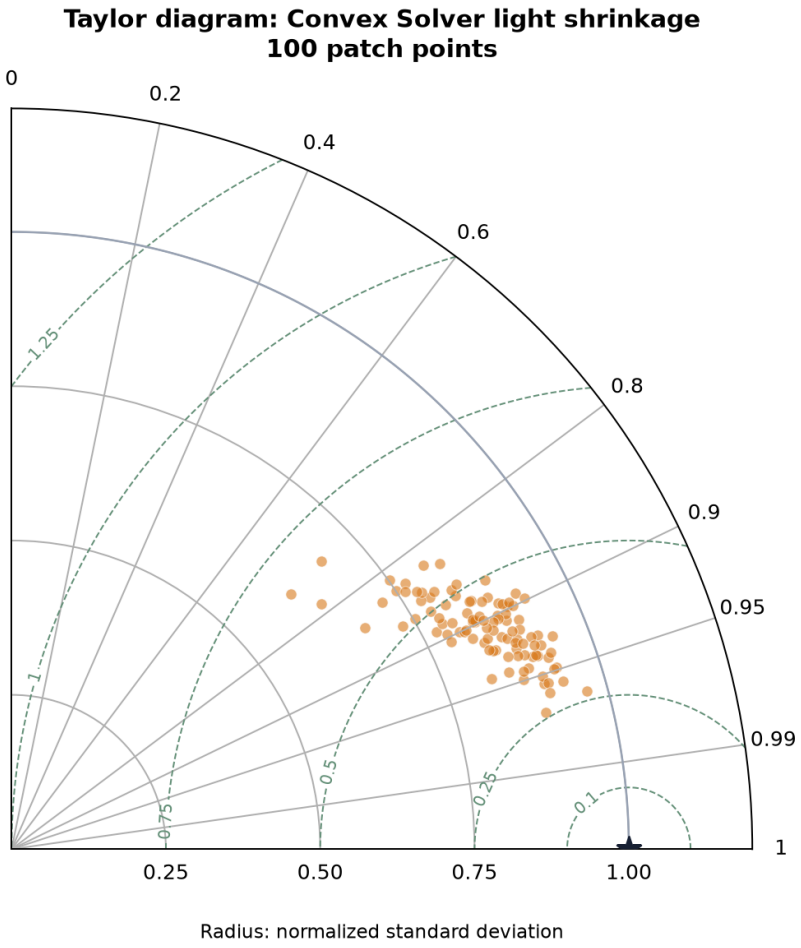
Solver	N	RMSE mean	RMSE min	RMSE max	Bias mean	Bias min	Bias max	Pearson mean	Pearson min	Pearson max
IDW	100	2.983	1.067	7.237	0.06152	-1.007	1.006	0.8056	0.6226	0.9347
ILDW	100	2.843	1.03	7.15	0.08054	-0.855	0.8596	0.8226	0.6294	0.9418
Convex Solver light shrinkage	100	2.161	0.8106	5.708	0.0697	-0.3811	0.5898	0.8982	0.732	0.9689
Convex Solver no shrinkage	100	2.161	0.8108	5.701	0.06996	-0.3865	0.5832	0.8982	0.7326	0.9685
Homotopy Solver light shrinkage	100	2.161	0.8106	5.707	0.06945	-0.3817	0.5886	0.8982	0.732	0.9689
Homotopy Solver no shrinkage	100	2.161	0.8109	5.701	0.06965	-0.3868	0.5825	0.8982	0.7326	0.9685
Solver(ILDW) light shrinkage	100	2.755	0.9785	7.147	0.0841	-0.8545	0.866	0.8342	0.6408	0.9595
Solver(ILDW) no shrinkage	100	2.76	0.9617	7.147	0.08137	-0.8554	0.8636	0.8346	0.6685	0.9515
ILDW-init Roy DCT	100	2.825	1.022	7.129	0.08265	-0.8434	0.8596	0.8253	0.6329	0.9432

Each row summarizes 100 patches. Mean/min/max are calculated separately across the per-patch RMSE, signed bias (prediction minus ground truth), and Pearson correlation. RMSE and bias are in mm/h. Only finite paired pixels are used; undefined correlations are excluded.

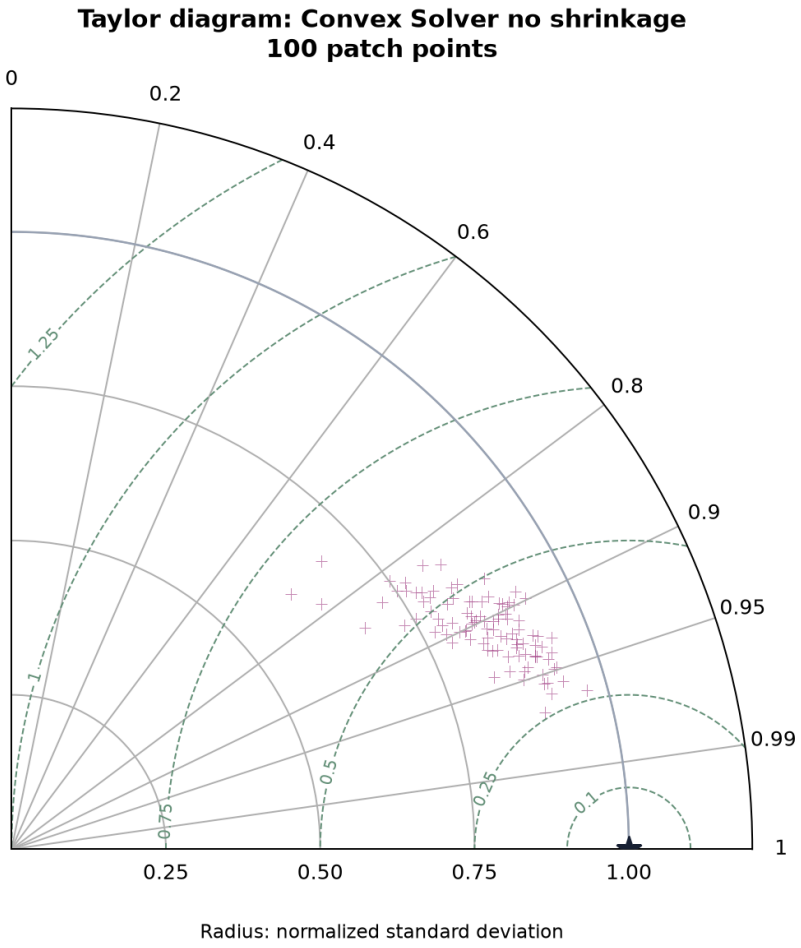
2. Taylor diagrams



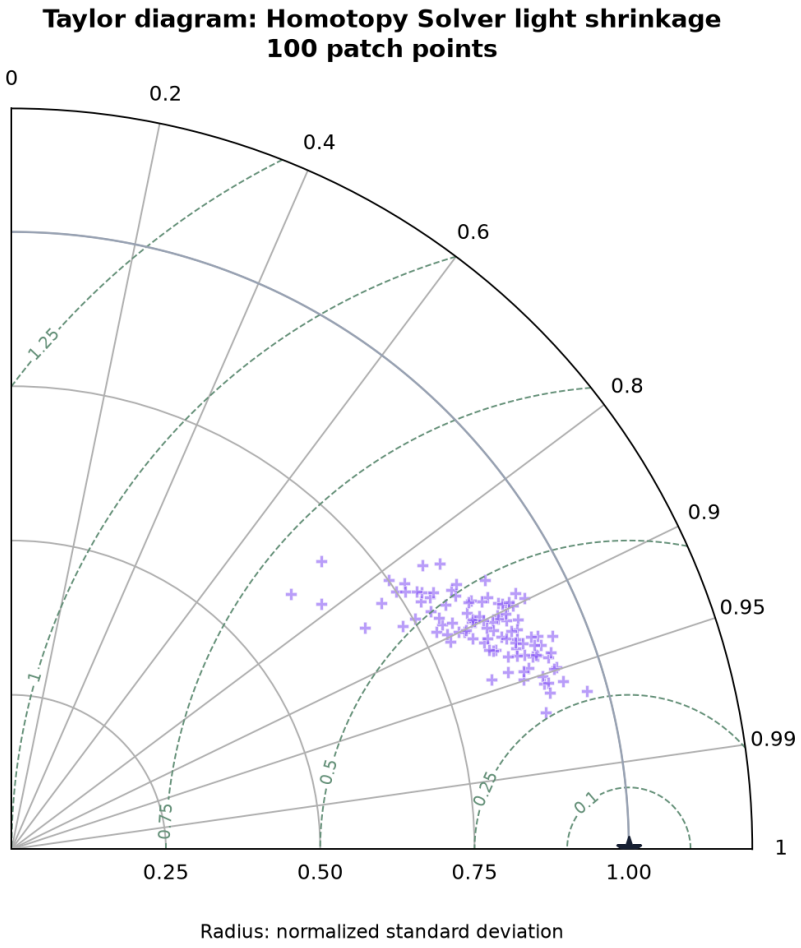
combined all solvers



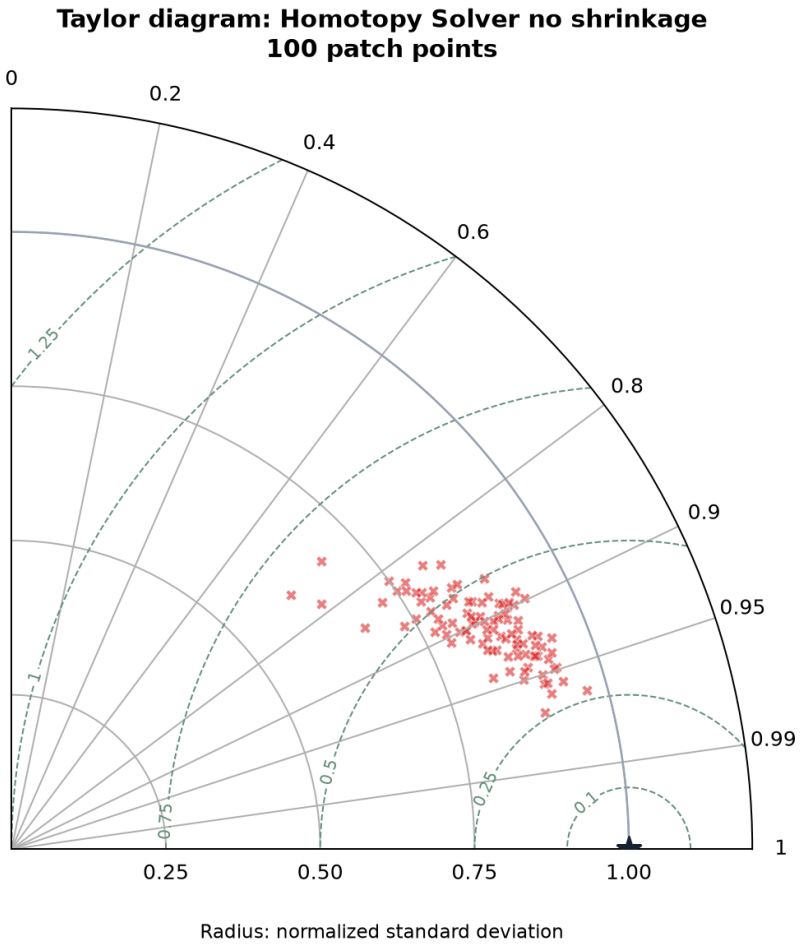
Convex Solver light shrinkage



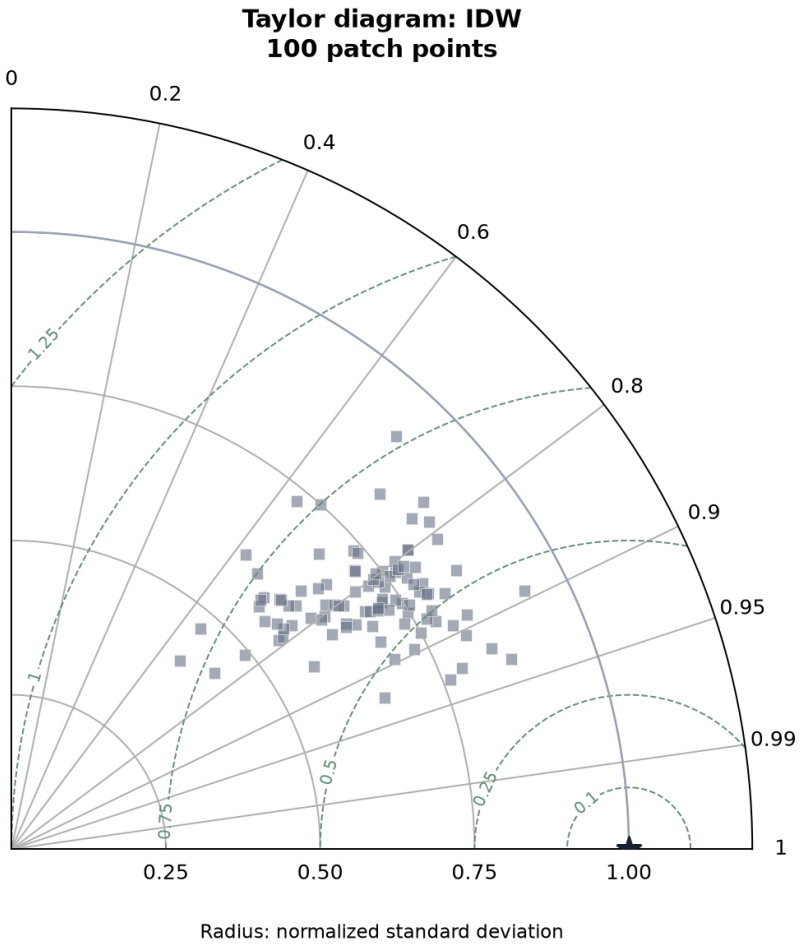
Convex Solver no shrinkage



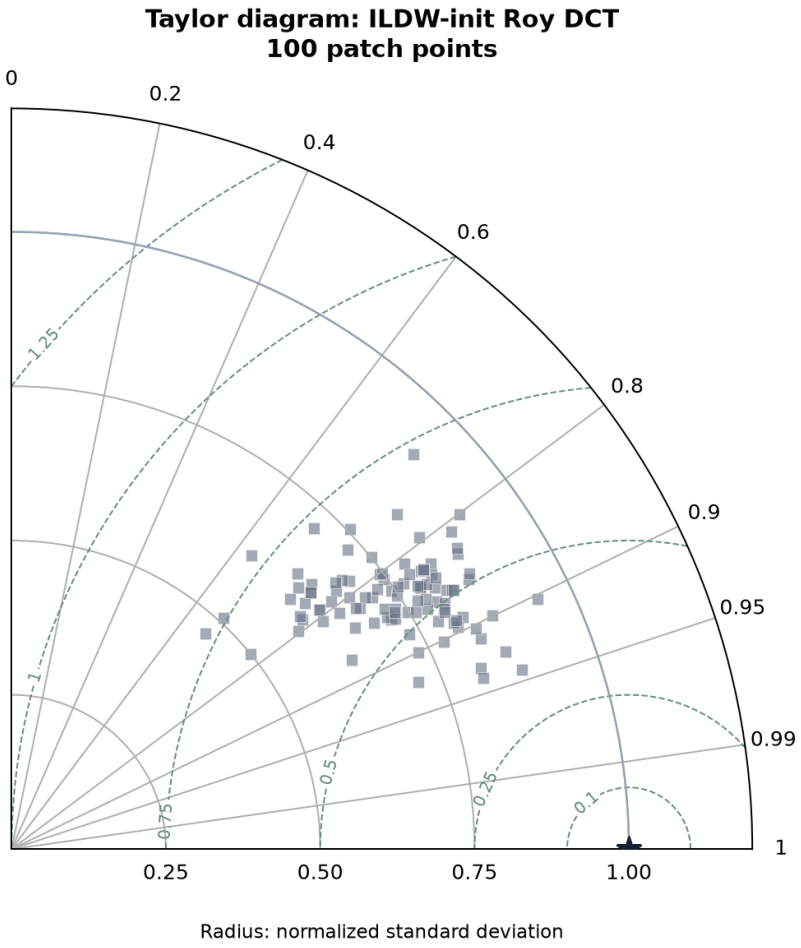
Homotopy Solver light shrinkage



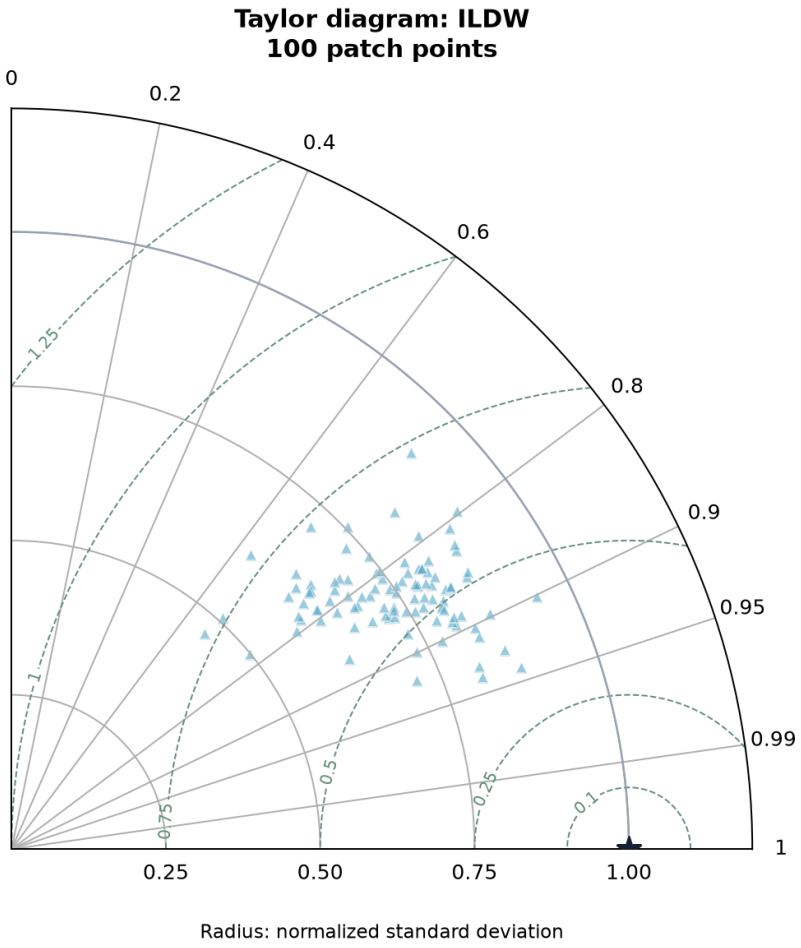
Homotopy Solver no shrinkage


Taylor diagram: IDW
100 patch points
0
0.2
0.4
0.6
0.8
0.9
0.95
0.99
1
1.25
1
0.75
0.5
0.25
0.1
0.25
0.50
0.75
1.00
Radius: normalized standard deviation

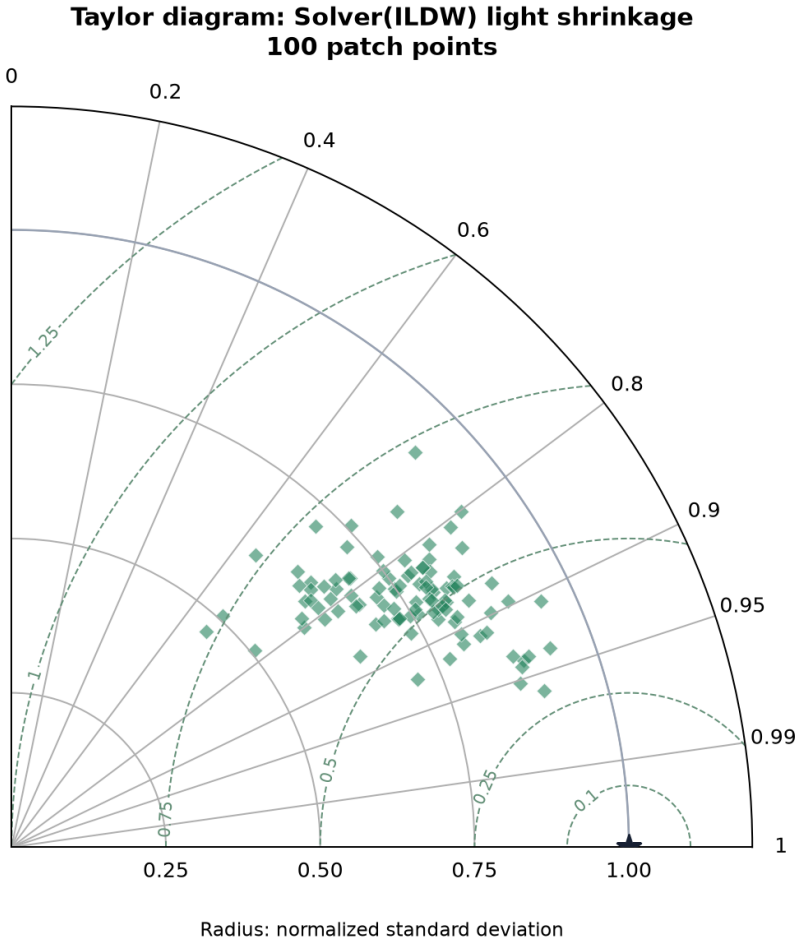
IDW



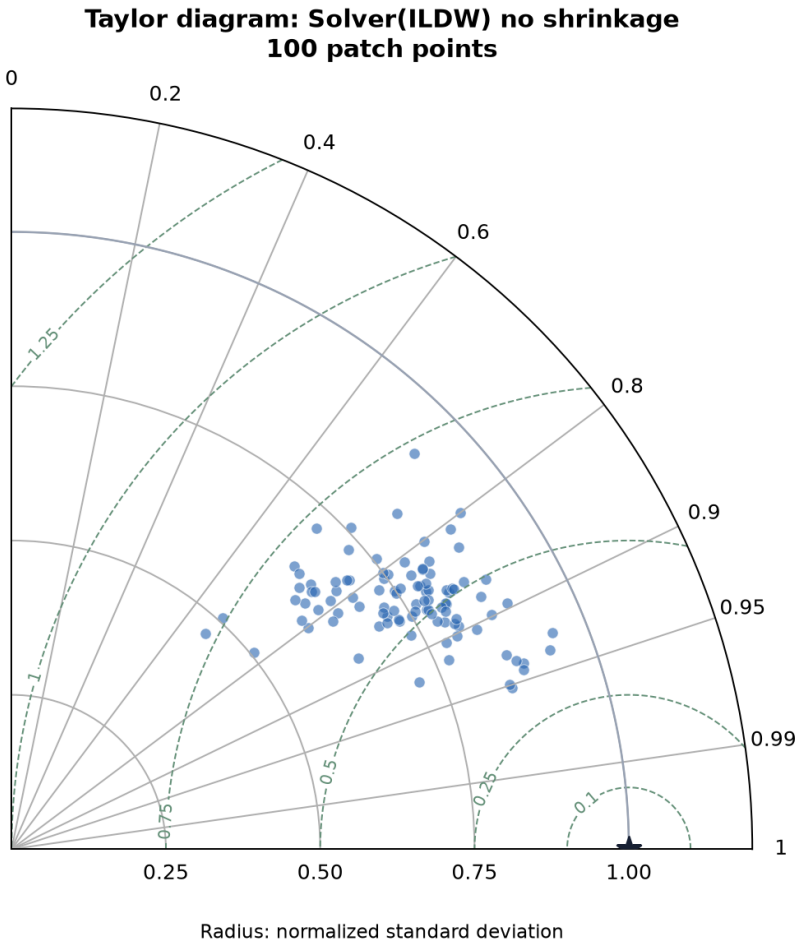
ILDW-init Roy DCT


Taylor diagram: ILDW
100 patch points
0
0.2
0.4
0.6
0.8
0.9
0.95
0.99
1
1.25
1
0.75
0.5
0.25
0.1
0.25
0.50
0.75
1.00
Radius: normalized standard deviation

ILDW



Solver ILDW light shrinkage



Solver ILDW no shrinkage

3. Wet/dry summary table

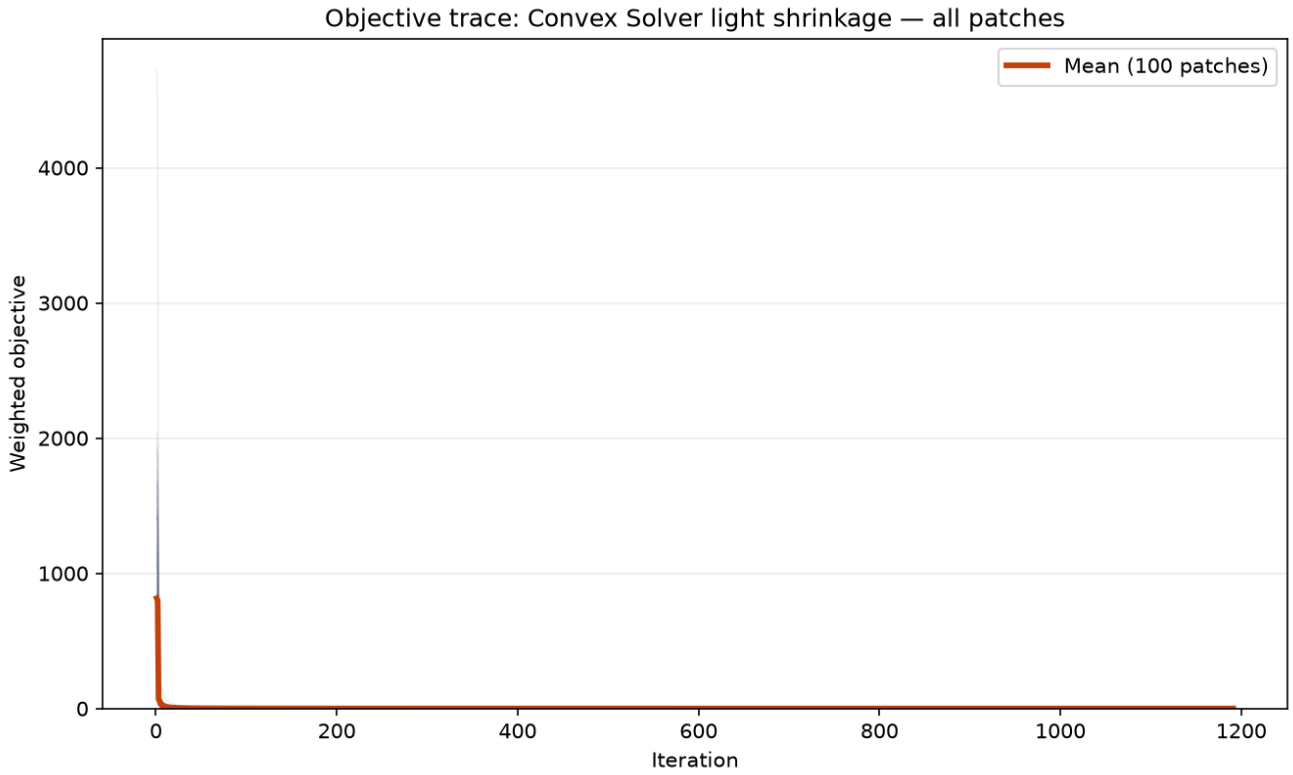
The wet/dry information is presented as a table rather than separate wet/dry images. Values are aggregated over all coverage bins and patches.

Intentional clutter note: the closely overlapping light-shrinkage and no-shrinkage traces are retained deliberately to show that the light-shrinkage term has negligible effect in this comparison.

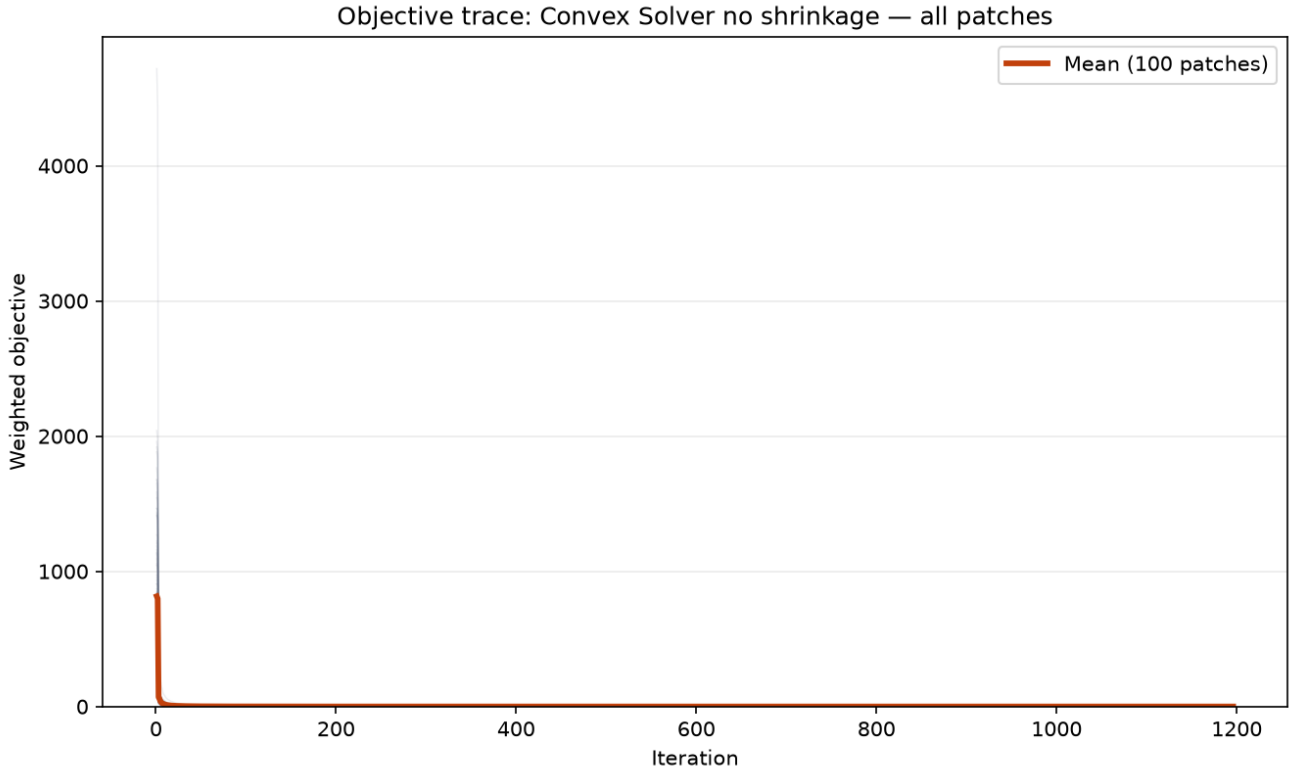
Solver	Mask	N pixels	Mean abs	Mean signed	Median abs
Convex Solver light shrinkage	rainy	20979281	0.3653	-0.1987	0.1888
Convex Solver light shrinkage	nonrainy	8605155	0.3842	0.3694	0.1004
Convex Solver no shrinkage	rainy	20979281	0.3656	-0.1988	0.189
Convex Solver no shrinkage	nonrainy	8605155	0.384	0.3692	0.1004
Homotopy Solver light shrinkage	rainy	20979281	0.3653	-0.1988	0.1888
Homotopy Solver light shrinkage	nonrainy	8605155	0.3842	0.3694	0.1002
Homotopy Solver no shrinkage	rainy	20979281	0.3655	-0.1989	0.1889
Homotopy Solver no shrinkage	nonrainy	8605155	0.384	0.3693	0.1002
IDW	rainy	20979281	0.5641	-0.3387	0.2966
IDW	nonrainy	8605155	0.8138	0.8063	0.3272
ILDW	rainy	20979281	0.5318	-0.315	0.2769
ILDW	nonrainy	8605155	0.741	0.7319	0.2754
Solver(ILDW)	rainy	20979281	0.5072	-0.3014	0.2602
Solver(ILDW)	nonrainy	8605155	0.7192	0.7094	0.2622

4. All-patch optimization behavior

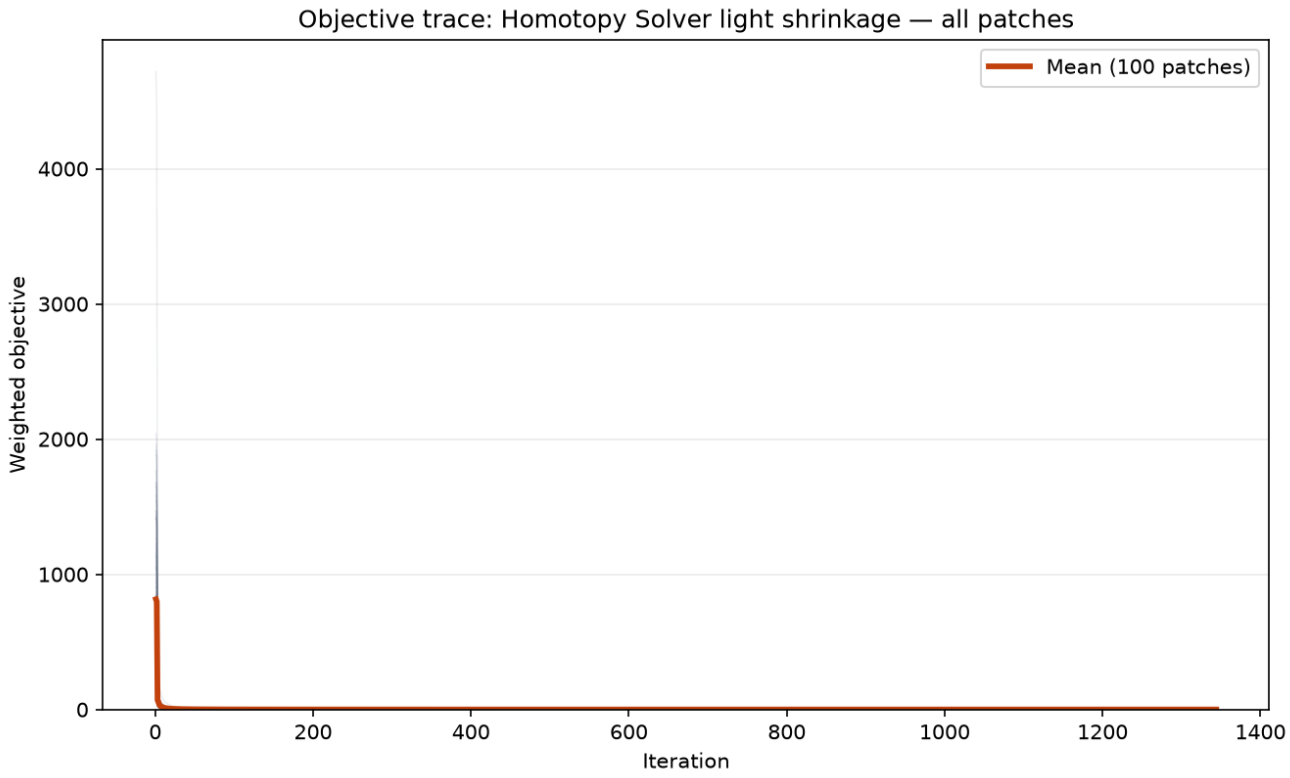
Faint lines are individual patches; the thick orange line is the arithmetic mean per iteration.



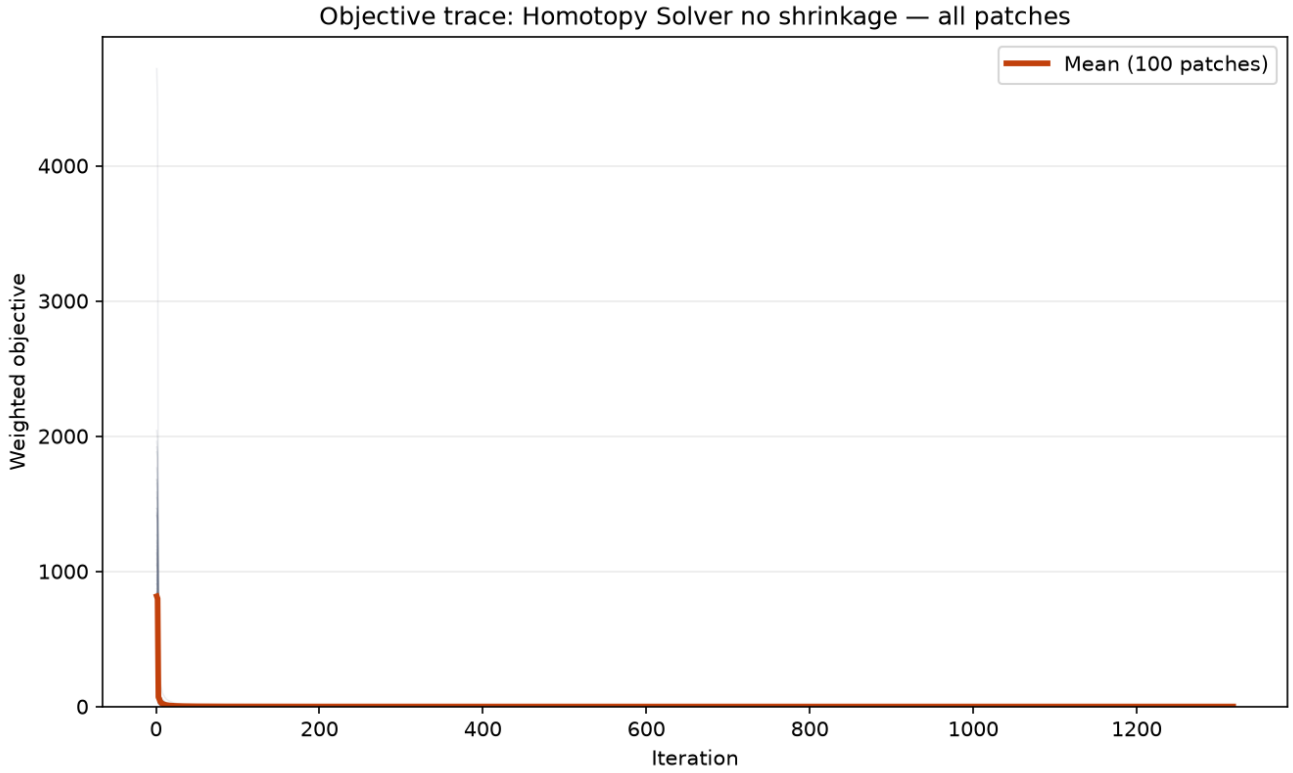
Convex Solver light shrinkage



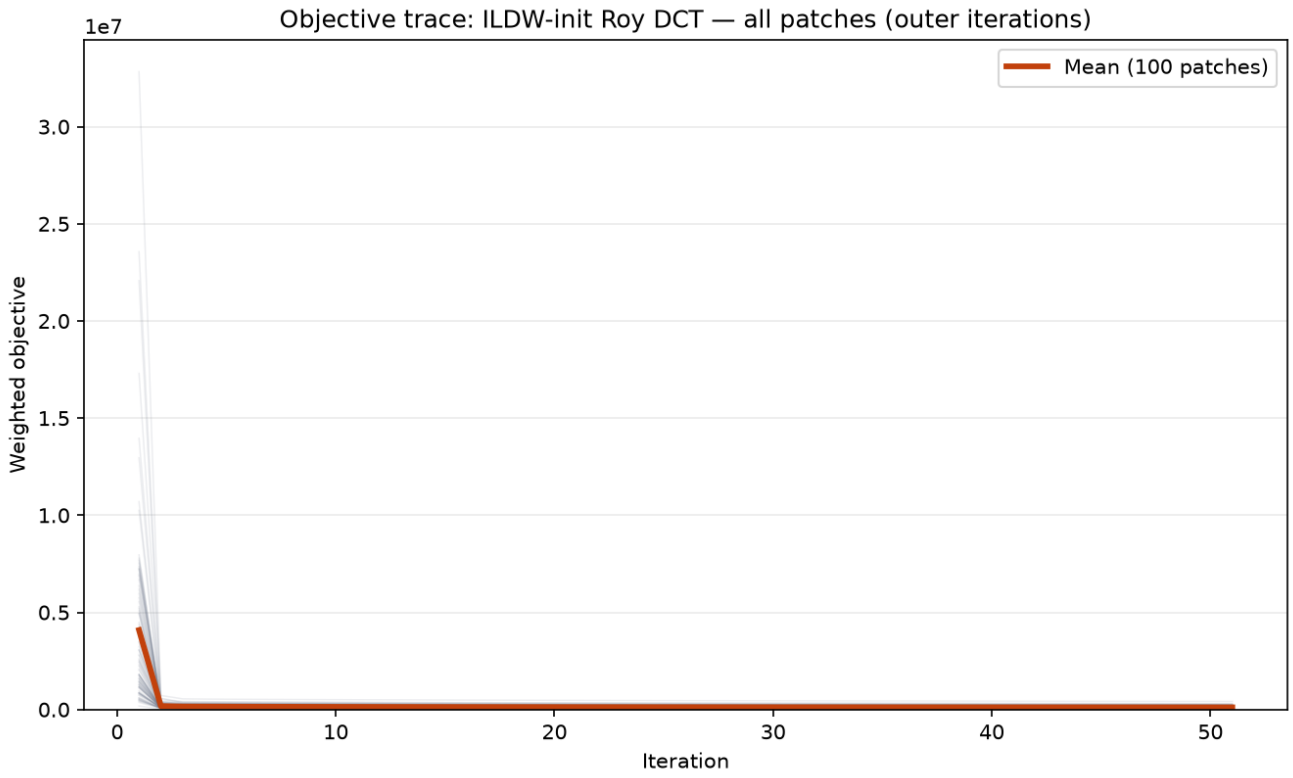
Convex Solver no shrinkage



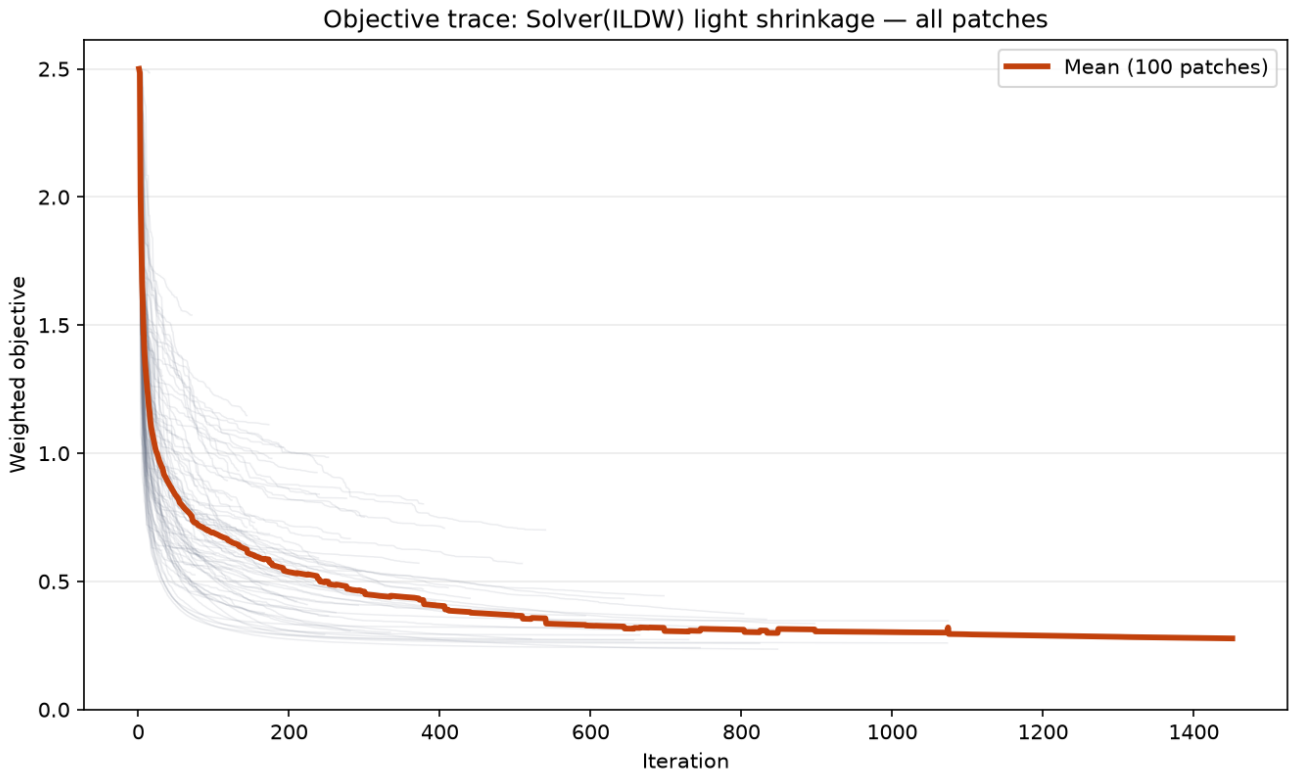
Homotopy Solver light shrinkage



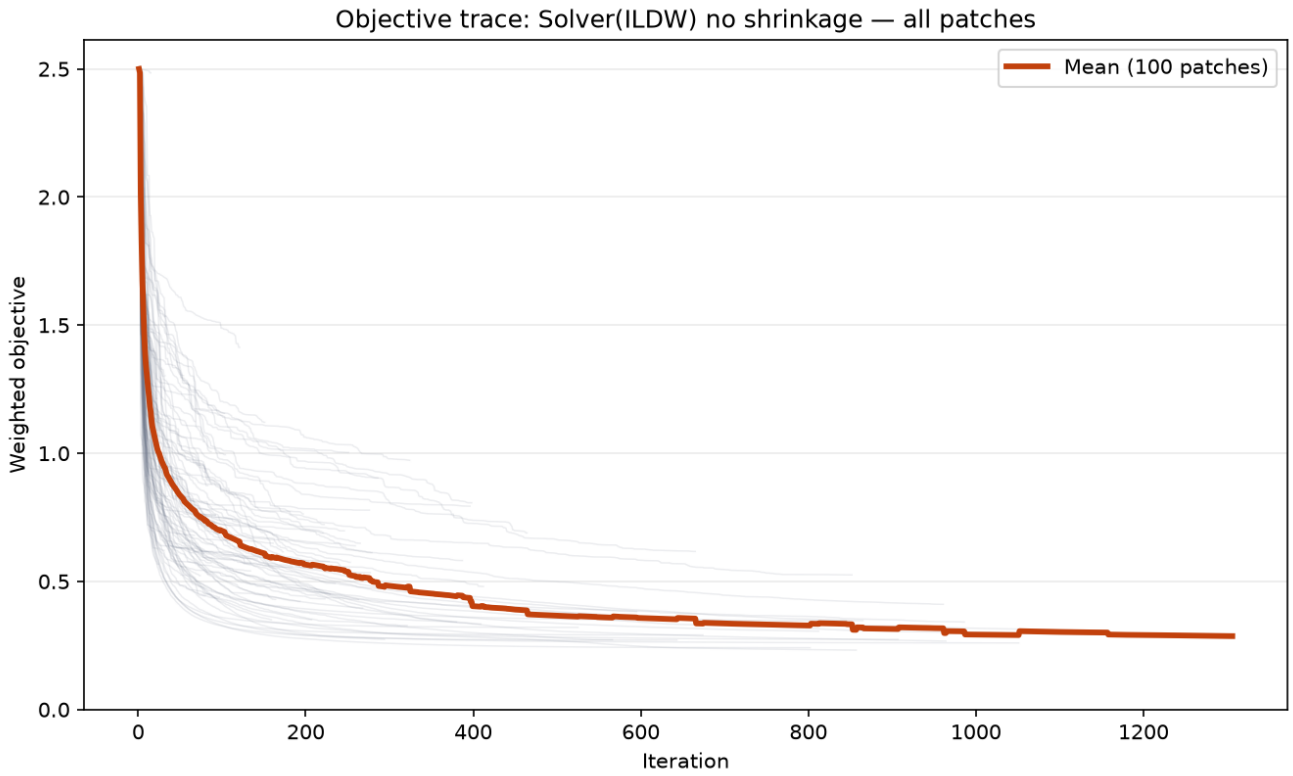
Homotopy Solver no shrinkage



ILDW-init Roy DCT



Solver ILDW light shrinkage



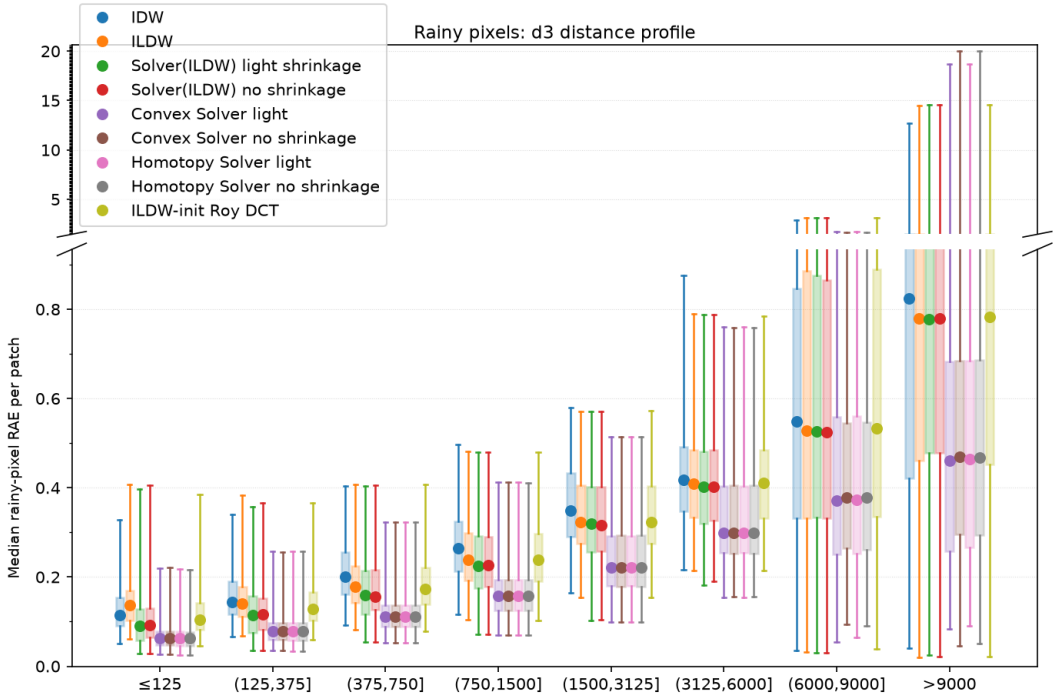
Solver ILDW no shrinkage

5. D3 distance summaries

Distance Profiles Box Whisker - d3 (third-closest link)



distance iqr medians nonrainy multi k3 box whisker



distance iqr medians rainy multi k3 box whisker

6. Solver runtimes

Solver directory	N	Mean s	Median s	Total h
sol_dir_convex_light_shrinkage	100	83.55	79.4	2.321
sol_dir_convex_no_shrinkage	100	79.96	73.71	2.221
ILDW-init Roy DCT	100	537.9	373.5	14.94
Solver(ILDW) no shrinkage	100	120.5	74.68	3.346
Solver(ILDW) light shrinkage	100	45.34	27.81	1.259